LiDAR Maze 10/12

Controls + mechanical/power

Project overview

- This is the second quarter of this project
- We have some code + designs from last quarter
- More hands-on engagement this quarter
- We are building a small “car” (~10x10 inches) to autonomously solve a maze
- We must build the car and the maze
- Will be using github

Raise of hands: new to this project?



The roof-mounted unit installed on Waymo's ride-hailing vehicles contains various hardware components for autonomous driving. *Courtesy of Waymo*

attendance

Miss one: warning

Miss two: speak to lead

Miss three: kicked from project

Please communicate your absences! We are flexible

Controls subteam

- Ensure that the robot follows movement controls precisely. If we tell it to move 10 inches east, it should do it exactly
- CAD a prototype of the chassis, do inverse kinematics (or some other math) to determine optimal mounting position of wheels, write motor control code

Leads: Pranav

Mechanical/power subteam

- CAD + 3D print robot chassis
- Design power delivery system from battery to all components
- Working with CAD softwares, LTspice, documentation, etc.

Leads: Andrew, Zane

Meeting today

- New members must complete a hands-on task to join their desired subteam
- Use the internet, AI, whatever you want to complete it
- PLEASE ask the leads questions (this isn’t a test! It shouldnt be silent in this room)
- This is to emulate how meetings will go. Leads are here to give you a task and help you, not be your tutor.
- You will have until the end of the meeting

task

Controls: wire arduino -> motor driver -> motor, get the motor moving

Mechanical: get “3-6V DC Miniatur brushed Toy Motor” from grabcad, import into onshape, design a 3d-printable enclosure to fasten the motor to a plate

Power: what do each of the pinouts on the buck converter/level shifter mean?

Ask your lead where to start!